

Not all differences matter. So who decides which ones do?

Human judgment in agentic-AI assessment of domain models — a literature review guided by a question

RESEARCH QUESTION

How can human judgment be captured, governed, and used to support agentic-AI-driven variability exploration in guideline operationalization scenarios, enabling reliable human–AI co-reasoning?

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Research-question wording is provisional, pending supervisor sign-off.

An AI can find every difference — not which ones matter

A deviation from a reference guideline is ambiguous by nature, and the ambiguity is where expertise is expensive.



A legitimate alternative

A different model can express the same domain meaning. Penalising it teaches the wrong lesson.



A genuine error

The model violates the domain, or encodes a misconception that should be corrected.



A defect in the guideline

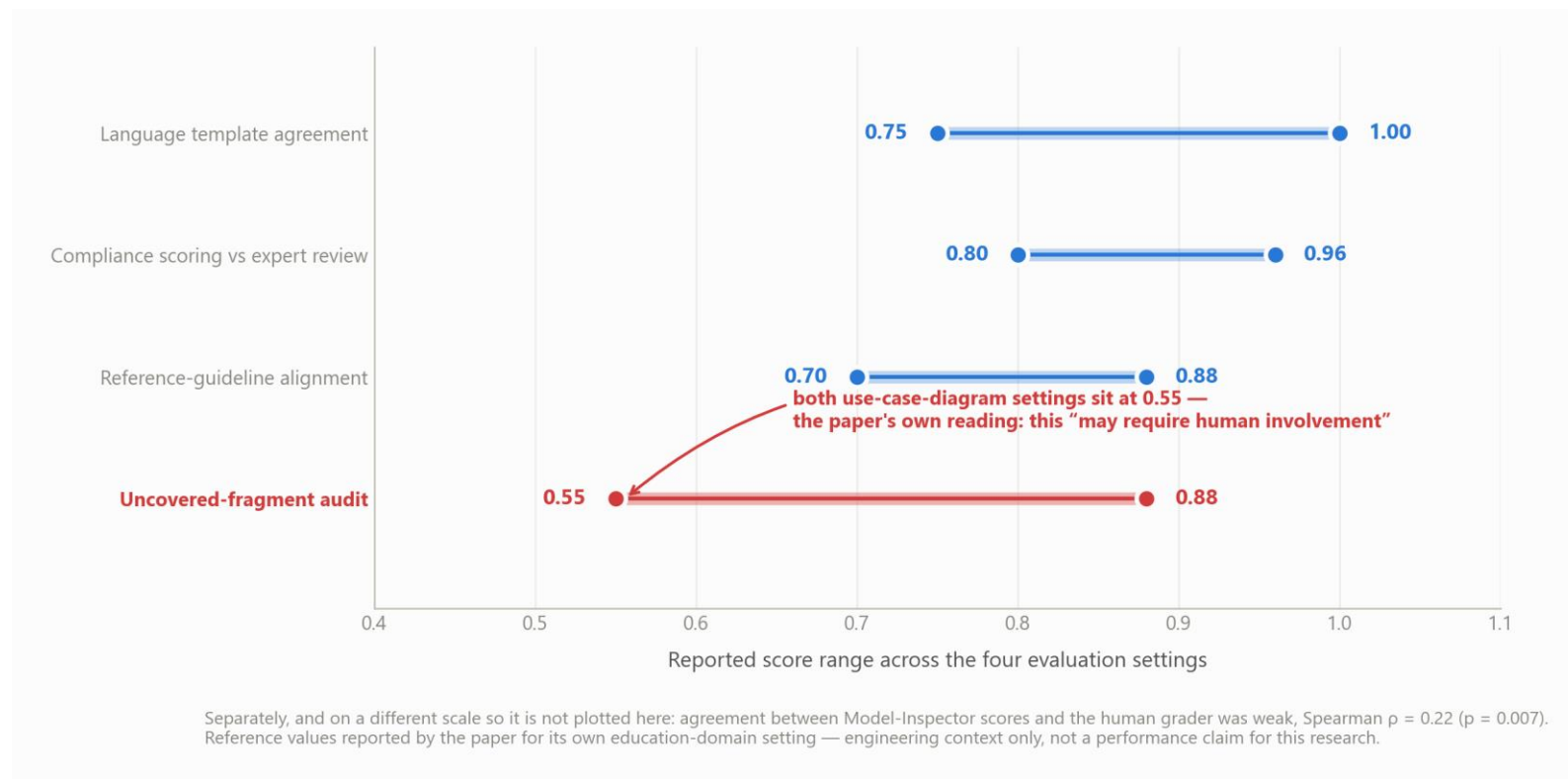
The reference itself is wrong or incomplete — the deviation is evidence against the norm.

Telling A, B and C apart is an interpretive act, not a classification.

It is exactly where automated assessment is least reliable — and where expert attention is scarcest.

The published framework names its own need for a human

VEGO-AI (Reinhartz-Berger, Bragilovski & Sturm, MODELS '26) assesses domain models with four coordinated LLM agents. It works — unevenly.



$$\rho = 0.22$$

Spearman correlation between the Model Inspector's compliance scores and the human grader ($p = 0.007$).

The paper's own future work:

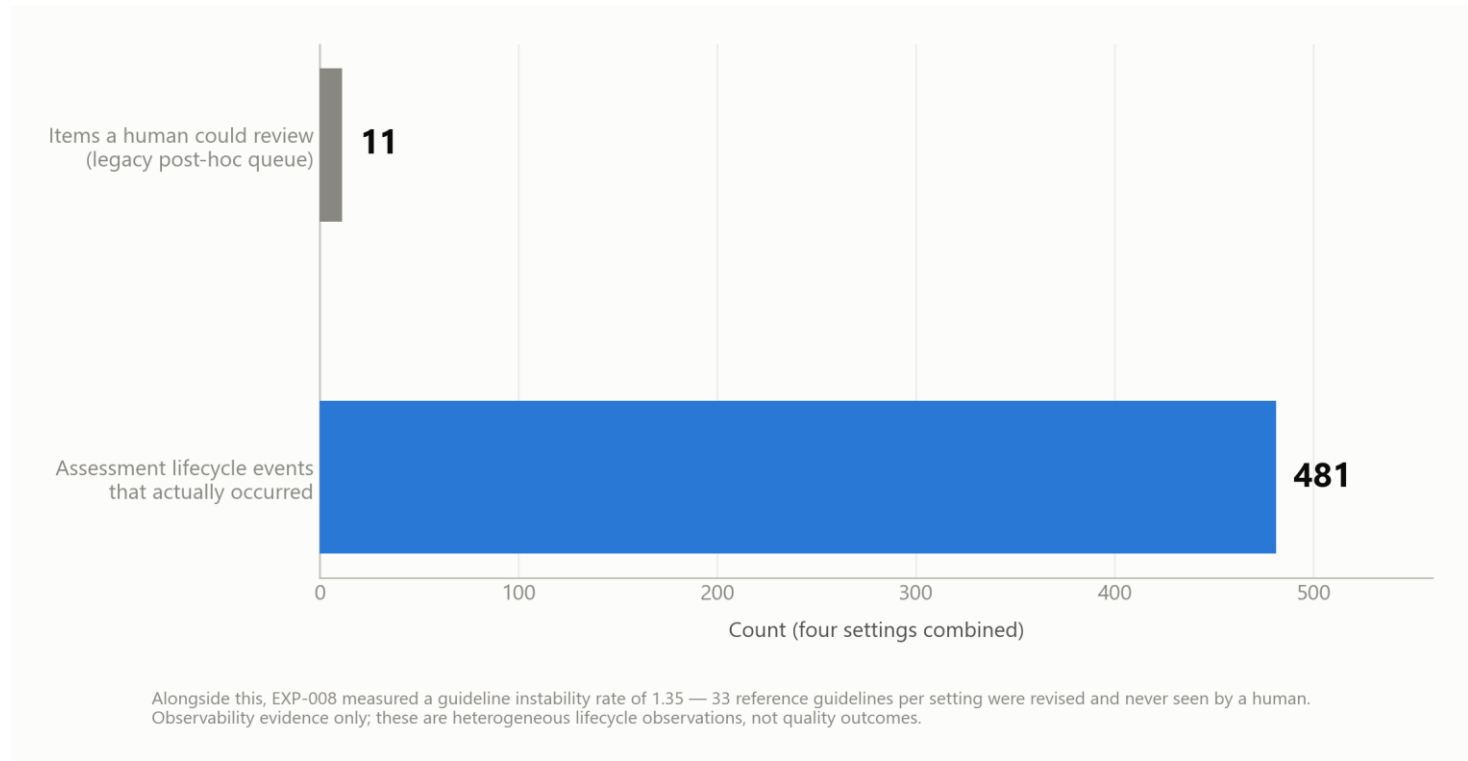
"incorporate human-in-the-loop oversight at key pipeline stages — guideline validation, compliance review, and variability classification"

The need for human judgment is not my assumption. It is a published, peer-reviewed finding — and it names the three stages.

Reference values reported by the paper for its own education-domain evaluation — engineering context, not a performance claim for this research.

I measured the same gap inside the system

Replaying the assessment pipeline as an event stream shows how little of its reasoning was ever reviewable.



44×

more lifecycle events occurred than the legacy queue could ever show

1.35

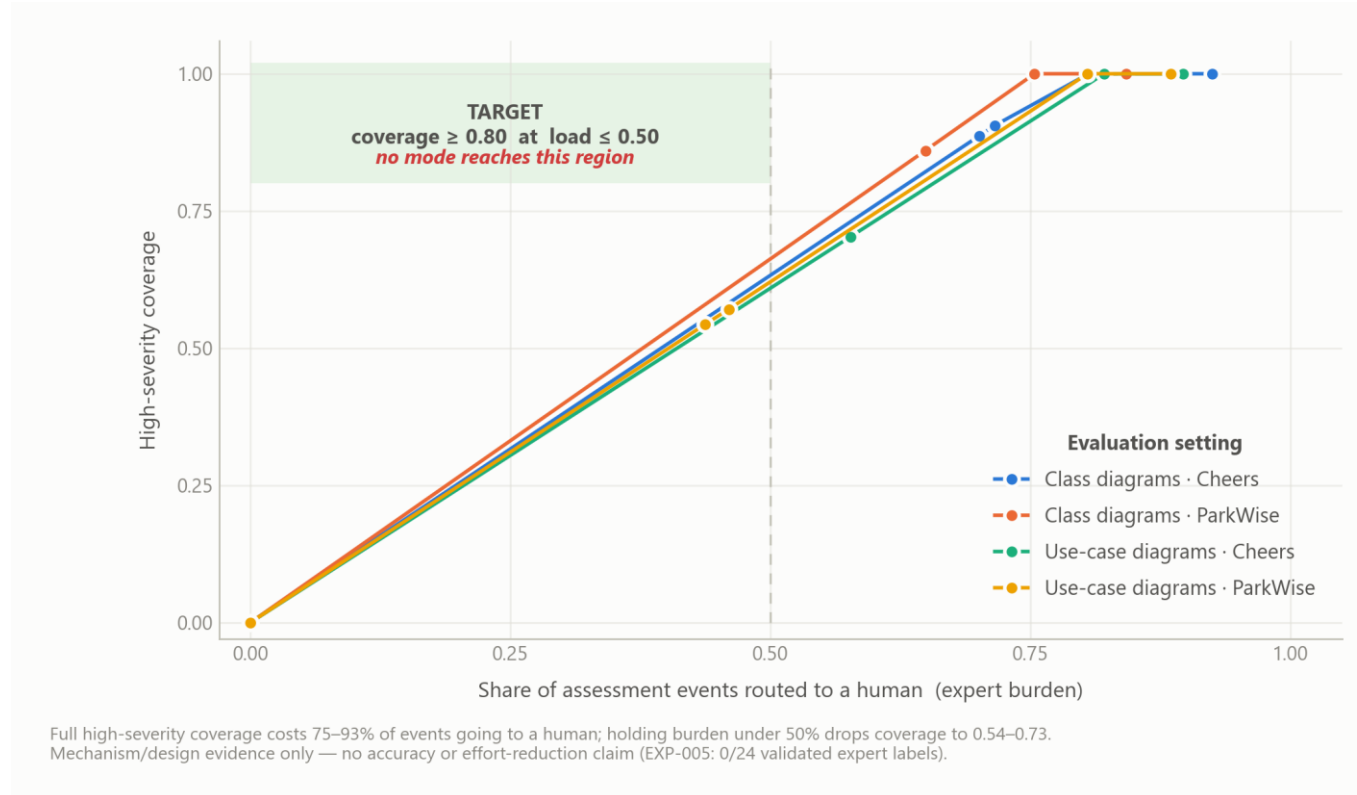
guideline instability rate — references were repeatedly rewritten

33

reference guidelines per setting revised with no human ever seeing them

... and simply asking the human more often does not solve it

Replaying five routing policies over those events maps the trade-off between expert burden and how much of the risk gets seen.



The dosage question

Review everything and automation is pointless. Review by a confidence threshold and you inherit the model's own miscalibration.

No policy reached 80% coverage of high-severity events while keeping expert burden under half.

That is SQ1, stated as a measurement rather than an opinion — and it is why “add a human in the loop” is not yet an answer.

The research question

How can human judgment be captured, governed, and used to support agentic-AI-driven variability exploration in guideline operationalization scenarios, enabling reliable human-AI co-reasoning?

It names the task, not the solution

The object of study is variability exploration in guideline operationalization — not a particular agent architecture.

Reuse is deliberately not in the headline

Whether captured judgment may be reused is a governed, testable concern for the sub-questions — not an assumption.

“Reliable” is the retained quality target

Earlier drafts also carried auditable, transferable and end-to-end; those are sub-question or artifact properties.

Three derived questions — which become three studies

Each sub-question owns one stage of the lifecycle: when to ask, what to keep, and where it may travel.

SQ1

Selective intervention

When and how should an agentic assessment system request human judgment, so that important uncertainties are addressed without unnecessary expert burden?

Study 1 · intervention architecture

SQ2

Governed knowledge reuse

How should expert judgment be represented, validated, reconciled and stored so it can be reused transparently, without unsafe generalization or loss of human authority?

Study 2 · judgment lifecycle

SQ3

Evaluation and transfer

How can reused judgment be evaluated and transferred across guideline-operationalization contexts — what generalizes, and what must adapt?

Study 3 · evaluation & transfer

How the review is being done

A structured, exploratory review: protocol first, then searching, screening, extraction and synthesis — recorded so it can be repeated.

Relevant research areas

Human involvement in agentic AI — the core any work placing human judgment inside a multi-step, tool-using AI process, whatever the domain

Guideline-operationalization scenarios assessment against an evolving reference: modelling guidelines, clinical guidelines

Knowledge capture, memory and reuse how a human decision is represented, stored, retrieved and governed

Evaluation and validity how human–AI assessment pipelines are evaluated, and what threatens the result

Deliberately excluded from the review body: enabling technologies (model architectures, local-inference tooling). They are engineering means, not the problem this review must justify.

Sources

Scopus · ACM Digital Library · IEEE Xplore · SpringerLink
Google Scholar and dblp for discovery, snowballing and verification.

Screening

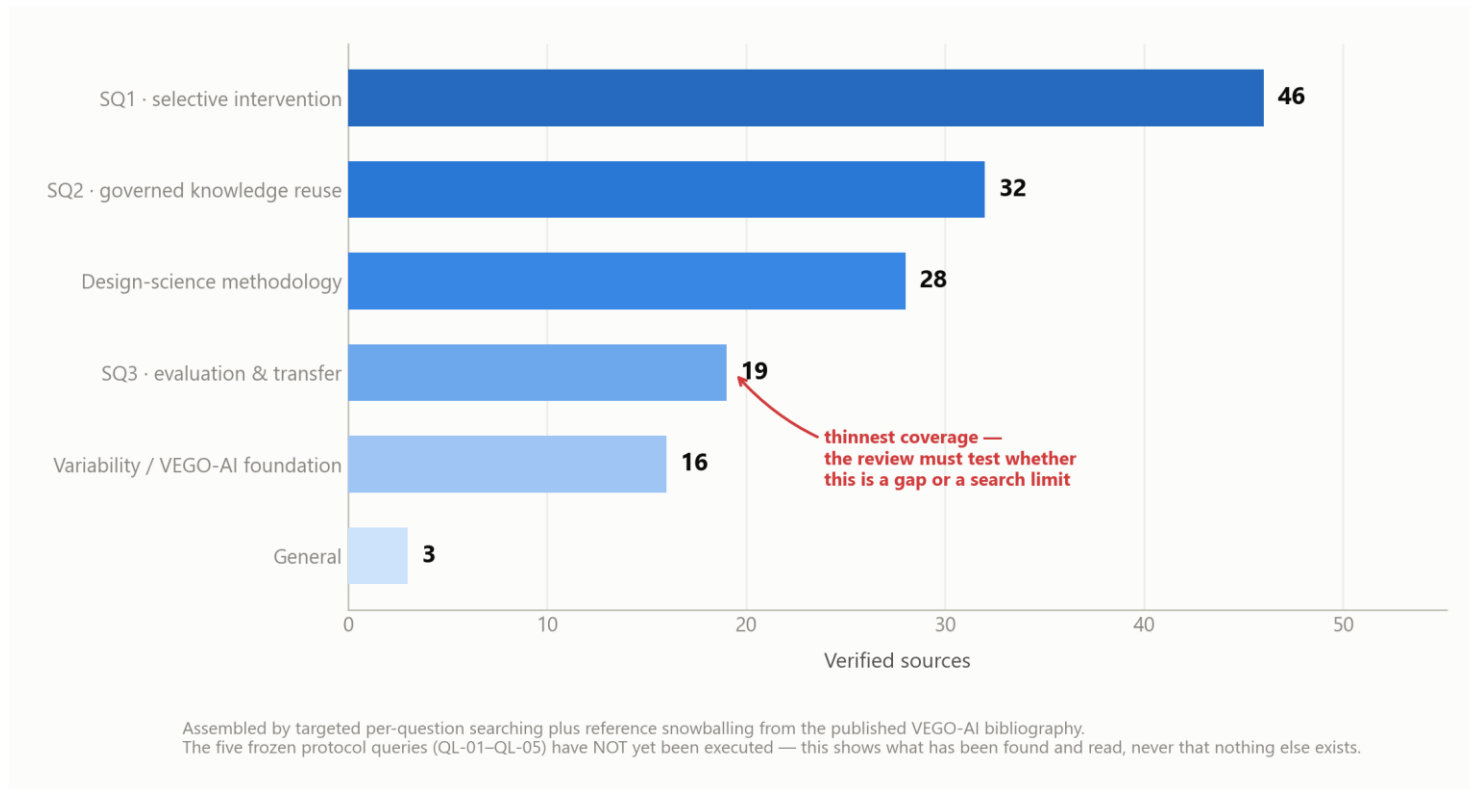
Include an explicit method, framework, artifact, empirical study or model.

Exclude passing mentions of “human in the loop”, product pieces, duplicates.

Every query, decision and exclusion reason is recorded.

What the search has produced so far

Each source was checked against an independent record before use — 132 of 144 confirmed, 11 partial, 1 quarantined.



Why verification came first

Generative tools invent plausible references. Every entry was checked against a publisher page, DOI, dblp or arXiv record; 1 that could not be confirmed is quarantined, not cited.

What this is not

The five frozen protocol queries have not yet been executed. This is what has been found and read — it is not evidence that nothing else exists.

Six streams — each answers a part, none the whole

Organised thematically, the reviewed corpus separates into six bodies of work, each stopping at a different point in the lifecycle.

A Escalation triggering & burden ~34

When to hand off: reject and defer rules, risk–coverage curves, uncertainty signals — plus evidence that human review is not free.

Leaves open in this work the expert's answer is consumed as a training label, decided per instance, and not carried forward.

B Judgment internalised in parameters 7

The paradigm this question is positioned against — learning from human preferences, model editing, unlearning — with its own critique.

Leaves open once absorbed into weights, a judgment has no record-level identity to inspect, attribute, scope or revoke.

C Externalised records & provenance ~17

Building blocks for holding judgment outside the weights: provenance standards, scope declarations, rationale schemas, disagreement as signal.

Leaves open in the reviewed work these sit at differing granularity and do not compose; provenance carries no validity or authority semantics.

D Runtime agent memory & reuse ~8

How a stored ruling could be reused at runtime — retrieval augmentation, memory taxonomies, self-reflection over stored records.

Leaves open the reviewed operations retrieve on similarity alone, with no authorisation, scope or expiry predicate.

E Model assessment & variability ~33

The application domain — automated and LLM-based assessment of conceptual models, and the vocabulary of planned, designed-in variation.

Leaves open the assessor's reasoning is consumed once; legitimacy of a difference is largely settled at construction time, not judged in context.

F Transfer, shift & validity ~43

How to state a transfer claim precisely and not fool yourself — shift vocabulary, paired in/out-of-context reporting, threats to validity.

Leaves open formulated over fitted models and static datasets; the reviewed set offers no category for a governed human-decision artefact.

Source counts are for the reviewed corpus only, and are approximate: a source may inform more than one stream. Each “leaves open” describes what the reviewed work does not cover — not a proven absence in the field.

The Judgment Lifecycle Grid

The unit of analysis is not a paper and not a topic — it is one expert ruling, and the five checkpoints it must pass between the moment an agent hesitates and the moment someone claims that ruling still holds elsewhere.

THREE CEILINGS

REUSE stops at L2

Nothing in the reviewed corpus checks a per-record scope gate before a stored ruling is applied to a new case. Reuse goes by similarity.

ASK stops at L2

The corpus shows that how a question is framed changes the answer — yet its measures score whether the system asked, never how.

RECORD is empty at L4

Reconciling conflicting rulings, and revoking one already applied, are not addressed in the reviewed corpus.

	TRIGGER what makes it stop and ask	ASK how the request is composed	RECORD what survives the expert's answer	REUSE what licenses applying it again	PROVE what evidence, obtained where
	L4 — not reached by any stream, on any checkpoint				
A · Escalation triggering, oversight & burden (~34)	L3	L2	L1	L0	L2
B · Judgment internalised in parameters (RLHF) (7)	absent	L1	L1	L0	L1
C · Externalised records: provenance, rationale (~17)	absent	L3	L3	L2	L1
D · Runtime agent memory & retrieval reuse (~8)	L0	absent	L2	L1	L1
E · Model assessment & variability engineering (~33)	L1	L2	L1	L1	L1
F · Transfer, shift & design-science validity (~43)	absent	absent	absent	L2	L3

Reading down a column shows the ceiling for that checkpoint. REUSE never passes L2 — no stream applies a machine-checkable, per-record scope gate before a stored ruling is reused. "absent" means the stream has no construct for that checkpoint at all. Positions describe the reviewed corpus only; the frozen searches are not yet executed.

Levels describe where the reviewed corpus reaches — they are not a claim that the field contains nothing higher. Executing the frozen protocol searches may raise any ceiling, and testing that is the next step.

What the framework makes visible

Each gap below is a thin or empty region of the matrix — stated as a boundary of the reviewed corpus, not as a proven absence.

Within the reviewed corpus —

G1	escalation and the judgment it obtains stay decoupled The corpus holds both halves and never joins them: intervention work consumes the expert's answer as a label and discards it; memory work supplies storage and retrieval with no authorisation or scope predicate.	SQ1 × SQ2
G2	intervention budgets are static, and coverage is not reported against reviewer load The formal budgets reviewed price a query at a constant, while the corpus's own clinical evidence shows cost rises with volume and repetition. The two quantities are never reported together.	SQ1 → SQ3
G3	triggers fire on content errors, not on “is this difference legitimate?” Whether a deviation is substantial variability or an occasional error is not itself the object of escalation in the reviewed work — yet it is precisely the judgment only a domain expert can make.	SQ1 × foundation
G4	expiry and revocation of a stored ruling are not addressed No reviewed source shows that a withdrawn ruling stops influencing later outputs, leaving no way to retire judgment that a later expert overturns.	SQ2

Gaps emerge from the findings, and are stated within the reviewed corpus. Executing the frozen protocol searches is the next step, and may confirm, narrow or remove any of them.

The gap this research takes on

G1 — the escalation–judgment loop is open.

The decision to ask a human, and the governance of what that human said, are developed as two separate literatures. This research closes the loop: an escalation trigger conditioned on a persistent, provenance-carrying, scope-bounded record of prior rulings — which writes back into it, so that what an expert once ruled changes which case is escalated next.

ENVIRONMENT

Expert review is scarce; the meaning of a deviation is contextual; and reference guidelines drift without human review.

ARTIFACT

A selective-intervention policy plus a governed, provenance-preserving judgment lifecycle around agentic assessment.

EVALUATION

Assessment quality, consistency, traceability, expert effort — then what transfers to a second guideline context.

What I cannot claim yet

No accuracy improvement, no reduction in expert effort, no safe generalization and no clinical performance. Those require independent expert labels (currently 0 of 24) and the executed searches — both are next, not done.

Where this goes next

1

Execute the frozen searches

Run the protocol queries per subsection, screen, and let the evidence confirm or correct these gaps.

2

Populate the extraction matrix

One row per included work, so the framework is filled by evidence rather than by expectation.

3

Turn each question into a study

Artifact and method per sub-question — the design-science structure this course sets out.

Thank you — questions welcome.

This work as design science

Design science requires bidirectional grounding in a real environment and a knowledge base, connected by an iterating design cycle (CL1–CL2).

RELEVANCE CYCLE

Environment

Expert review is scarce and scarce attention must be spent well; deviation meaning is contextual; reference guidelines drift unseen (EXP-006/008).

DESIGN CYCLE

Build → evaluate → iterate

An escalation trigger conditioned on a persistent, scope-bounded judgment record (G1) — built and evaluated per sub-question, per the study contract.

RIGOR CYCLE

Knowledge base

144 verified sources organised into six streams and the Judgment Lifecycle Grid — itself a candidate addition back to the knowledge base.

Design-science guidelines checklist

How the seven guidelines map onto this research, stated plainly rather than assumed.

1	Design as an artifact	An escalation-trigger policy and a governed judgment lifecycle — a method and a model, in Hevner's terms.
2	Problem relevance	Motivated by the published framework's own reported need for human judgment ($p = 0.22$) and by measured mechanism gaps in this project (EXP-006–008).
3	Design evaluation	Per-study measures already specified (SQ1–SQ3); accuracy/effort/generalization claims explicitly excluded until independently evidenced.
4	Research contributions	The Judgment Lifecycle Grid is a candidate contribution to the knowledge base, not only an analysis tool for this review.
5	Research rigor	144 sources, each independently verified against a publisher record, DOI, dblp or arXiv — zero fabricated citations.
6	Design as a search process	Six candidate gaps narrowed to one by evidence; the frozen protocol searches are the next iteration, not a one-shot search.
7	Communication of research	This seminar presentation for a technical audience; three planned publications per the study contract for the research community.

Use of generative AI in preparing this work

Declared per the course requirement that the work state whether and how GenAI was used, and for what.

Used for

- Brainstorming and refining the wording of the research questions.
- Discovering candidate literature and drafting search-query families.
- Screening support: summarising abstracts to triage relevance.
- Extracting structured fields from sources into the review workbook.
- Drafting and rephrasing slide text, and generating the figures from measured result files.

Not used for

- Generating citations. Every reference was verified against a publisher page, DOI, dblp or arXiv record; unverifiable entries are quarantined.
- Deciding the research questions or the message of the review.
- Designing the analysis framework — that is the conceptual work.
- Determining gaps. A model reports what exists, not what is missing; gaps come from the framework and remain provisional until the searches run.

Evidence boundary held throughout

Separating what the current work can demonstrate from what still requires independent evidence.

Can be stated (mechanism / observability)

- 481 lifecycle events reconstructed against a legacy queue of 11.
- Guideline instability measured at 1.35; 33 guidelines per setting never reviewed.
- Coverage-versus-burden frontier mapped across five routing policies.
- The protected baseline is unchanged — 0 of 27 classifications altered.

Not claimed — requires evidence that does not exist yet

- No claim of improved assessment accuracy — 0 of 24 generalization-safe expert labels are validated.
- No claim of reduced expert workload.
- No claim of safe generalization or transfer across contexts.
- No claim of any kind about clinical performance.
- No claim that the literature holds no prior solution — the protocol searches are not yet executed.